

Intelligent Energy Management for Hospital Microgrids

A Hierarchical Model Predictive Control Approach

23AID305: Control Systems

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Introduction & Motivation

Why Focus on Hospitals?

- **High Energy Demand:** Hospitals operate 24/7 with energy-intensive equipment (MRI, HVAC), making them major energy consumers.
- **Critical Loads:** Uninterrupted power is a matter of life and death. Reliability and resilience are paramount.

Complex Challenges:

- **Volatile Costs:** Time-of-Use (TOU) electricity pricing requires smart scheduling.
- **Intermittent Renewables:** Integration of solar PV introduces uncertainty.
- **Dynamic Loads:** Unpredictable, high-power loads and complex thermal dynamics.

Our Solution: A proactive, two-level Model Predictive Control (MPC) system to intelligently manage these complexities.

Components of the Hospital Microgrid:

- **Generation:** Utility Grid, Solar PV, Diesel Generator (DG)
- **Storage:**
 - **BESS:** For price arbitrage and load shifting.
 - **SC:** For power smoothing and rapid response.
- **Loads:**
 - Controllable: HVAC system.
 - Uncontrollable: Lighting, medical equipment.

Energy Management System (EMS): Coordinates all assets for cost-efficient, reliable operation.

System Modeling: The Physics

1. Thermal Zone Model (RC Analogy)

$$\frac{dT_{in}}{dt} = \frac{1}{C} \left(\frac{T_{out} - T_{in}}{R} + Q_{internal} + Q_{hvac} \right)$$

- C : Thermal mass (kWh/°C)
- R : Wall thermal resistance (°C/kW)

2. BESS Model

$$\frac{d(SOC)}{dt} = - \frac{P_{bess}(t) \cdot \eta \operatorname{sgn}(-P_{bess})}{E_{capacity}}$$

- P_{bess} : Battery power (positive = discharge)
- $E_{capacity}$: Battery capacity (kWh)

The Hierarchical MPC Strategy

Why Two Levels? Separation by timescale.

Level 1: Supervisory MPC (Planner)

- Horizon: 24 hours, Timestep: 15 min.
- Objective: Economic optimization — minimize total cost.
- Output: Optimal 24-hour plan for Grid, DG, BESS, HVAC.

Level 2: Real-Time MPC (Reactor)

- Horizon: Instantaneous, Timestep: 1 min.
- Goal: Track plan, reject disturbances.
- Logic: Corrects deviation between planned and actual net power.

Level 1: Objective Function & Constraints

Objective:

$$\min_u J = \sum_{k=0}^{N-1} (C_{grid,k} P_{grid,k} + C_{dg,k} + C_{degradation,k}) + \text{Penalty}_{thermal}$$

Constraints:

- **Power Balance:**

$$P_{PV} + P_{Grid} + P_{DG} + P_{BESS} - P_{Load} - P_{HVAC} = 0$$

- **System Dynamics:**

$$x_{k+1} = Ax_k + Bu_k + Dd_k$$

- **Operational Limits:**

$$0 \leq SOC_{BESS} \leq 1, \quad |P_{BESS}| \leq P_{BESS,max}$$

Level 1: State-Space MPC Formulation

$$x_{k+1} = Ax_k + Bu_k + Dd_k$$

State Vector:

$$x_k = [T_{in,k}, SOC_{BESS,k}]^T$$

Control Vector:

$$u_k = [P_{grid,k}, P_{dg,k}, P_{bess,k}, Q_{hvac,k}]^T$$

Disturbance Vector:

$$d_k = [T_{out,k}, Q_{internal,k}]^T$$

Level 1: Objective Function & Constraints

Objective:

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Constraints:

- **Power Balance:**

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- **System Dynamics:**

$$x_{k+1} = Ax_k + Bu_k + Dd_k$$

- **Operational Limits:**

$$0 \leq SOC_{BESS} \leq 1, \quad |P_{BESS}| \leq P_{BESS,max}$$

Deriving the State-Space Model

Step 1: Discretize the Thermal Model

$$T_{in,k+1} = T_{in,k} + \frac{\Delta t}{C} \left(\frac{T_{out,k} - T_{in,k}}{R} + Q_{internal,k} + Q_{hvac,k} \right)$$

Step 2: Discretize the BESS Model

$$SOC_{k+1} = SOC_k - \frac{\Delta t}{E_{capacity}} P_{bess,k}$$

Step 3: Assemble State-Space Form

$$x_{k+1} = Ax_k + Bu_k + Dd_k$$

$$A = \begin{bmatrix} 1 - \frac{\Delta t}{RC} & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} \frac{\Delta t}{C} & 0 \\ 0 & -\frac{\Delta t}{E_{capacity}} \end{bmatrix}, \quad D = \begin{bmatrix} \frac{\Delta t}{RC} & \frac{\Delta t}{C} \\ 0 & 0 \end{bmatrix}$$

Level 2: Real-Time Disturbance Rejection

Handling Reality

- Forecasts have errors and disturbances.
- Real-Time Logic:

$$P_{error} = P_{net,measured} - P_{net,planned}$$

- **Correction Strategy:**

- 1 SC handles fast spikes:

$$P_{SC,correction} = \text{clip}(P_{error}, -P_{SC,max}, P_{SC,max})$$

- 2 BESS handles slower residual:

$$P_{BESS,correction} = \text{clip}(P_{error} - P_{SC,correction}, -P_{BESS,max}, P_{BESS,max})$$

- 3 Grid compensates final mismatch.

The Brains of Proactive Control

- Accurate forecasts of Temperature, Load, PV Power are vital for MPC optimization.
- Uses a data-driven approach with a simple ANN (ForecastANN class):
 - **Input:** Hour of the day.
 - **Network:** 2 hidden layers (128 neurons, ReLU activation).
 - **Output:**
 - Outdoor Temperature (T_{out})
 - PV Generation (P_{PV})
 - Uncontrollable Load (P_{Load})
- **Training:** On synthetic data with normalization (MinMaxScaler) and Mean Squared Error loss.

The Hybrid Energy Storage System (HESS)

To manage energy effectively across different timescales, we combine:

BESS (Battery Energy Storage System) – The Marathon Runner

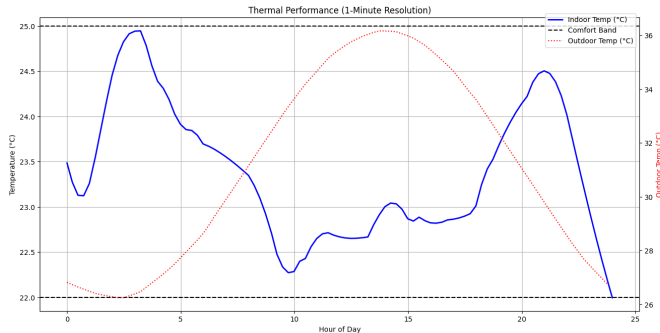
- High energy density, ideal for energy arbitrage.
- Managed by the supervisory MPC.
- Limitation: Slower response; degrades faster with frequent cycling.

Supercapacitor (SC) – The Sprinter

- High power density; reacts within milliseconds.
- Absorbs fast transients (e.g., equipment spikes).
- Limitation: Low energy capacity.

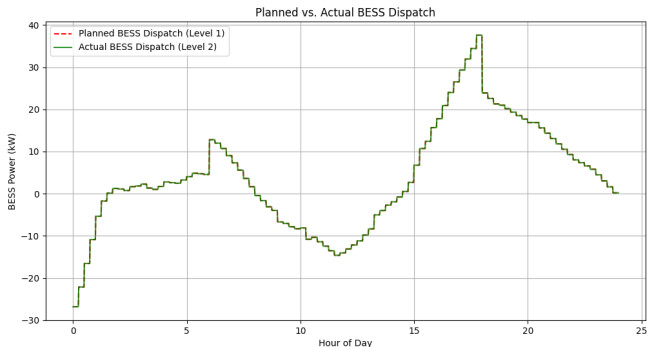
Synergy: BESS handles slow, long-term power planning, while SC manages fast disturbances, extending BESS life and system stability.

Result Plot 1: Thermal Performance



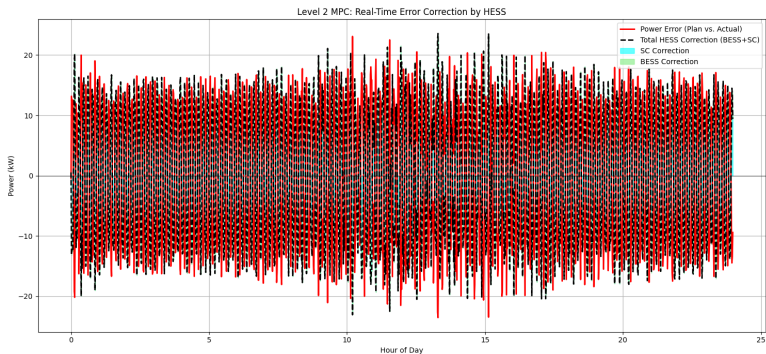
The MPC maintains indoor temperature perfectly within the 22°C–25°C comfort band while adapting to outdoor fluctuations.

Result Plot 2: Planned vs Actual BESS Dispatch



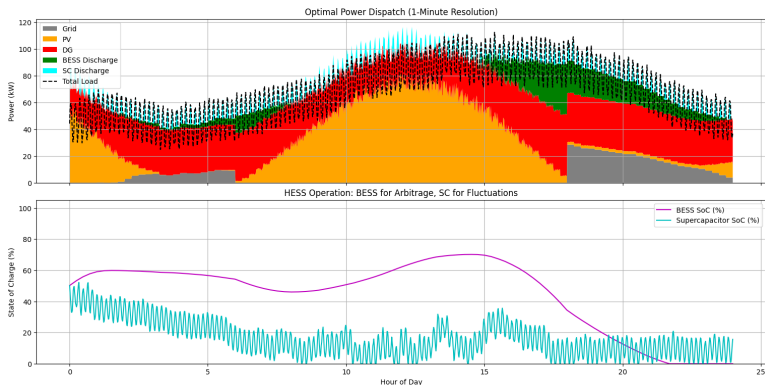
The real-time controller accurately tracks the planned BESS schedule, proving precise coordination between Level 1 and Level 2 MPC.

Result Plot 3: Level 2 MPC Correction



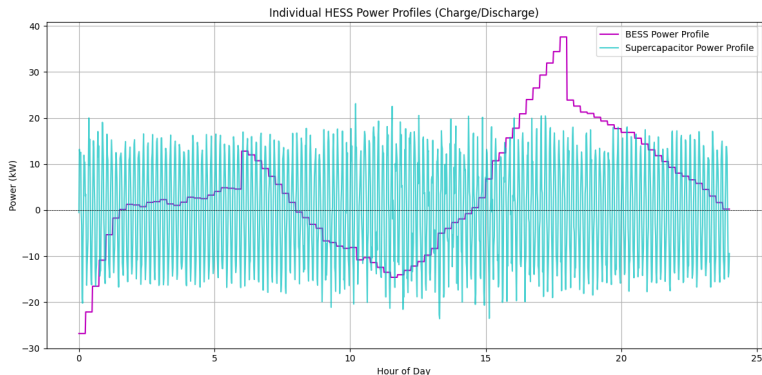
The Level 2 MPC effectively cancels high-frequency power errors, ensuring grid stability through real-time corrective action.

Result Plot 4: Optimal Power HESS Operation



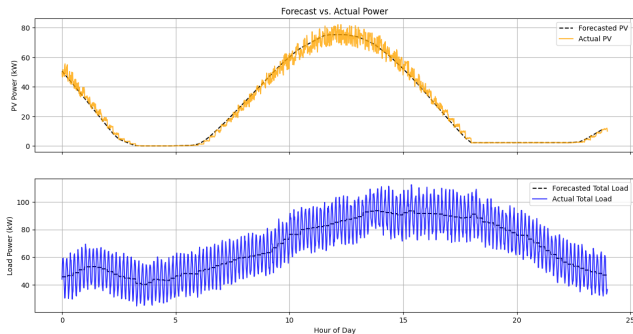
MPC optimally coordinates PV, Grid, and BESS power flows while eliminating Diesel Generator usage for cost efficiency.

Result Plot 5: HESS Power Profiles



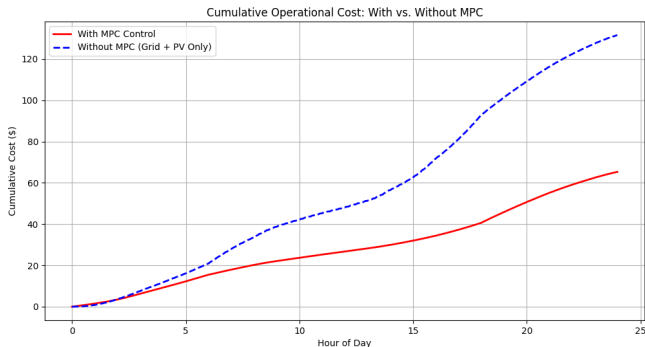
BESS performs smooth energy balancing while the Supercapacitor handles fast transients — demonstrating hybrid synergy.

Result Plot 6: Forecast Plots



Forecasts accurately capture daily PV and load trends, enabling proactive MPC planning and robust real-time control.

Result Plot 7: Cumulative Cost



The MPC cuts total operational cost by about 50%, achieving major savings during peak electricity price hours.

Conclusion

- Developed and simulated a hierarchical MPC for hospital microgrids.
- Achieved cost minimization, thermal comfort, and reliability.
- The two-level design ensures both long-term planning and real-time adaptation.

Future Work

- Train ANN on real hospital data.
- Extend models to multi-zone buildings.
- Introduce Stochastic MPC for uncertainty handling.
- Implement Hardware-in-the-Loop (HIL) validation.